

A quadratic precision cutoff for approximately fair dice

Research proof; three independent full mathematical reviews completed

30 September 2026

Status. Three independent full mathematical reviews completed with no gap identified; this is not formal verification. All constants below are absolute. Write $\log(1/0) = +\infty$.

Theorem 1. *Suppose $n \geq 3$ independent equiprobable finite dice have no cross-die ties, and every strict order has probability in $[(1-\varepsilon)/n!, (1+\varepsilon)/n!]$, where $0 \leq \varepsilon \leq 1/2$. Every die has at least*

$$cn \min\{n, \log(1/\varepsilon)\}$$

faces, for an absolute $c > 0$.

We prove the following tensor version. Let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing, with positive probability row weights w , and suppose all complete failure patterns satisfy

$$\mathbb{E}_w \prod_{j \in F} (1 - p_j) \prod_{j \notin F} p_j \in \left[\frac{1 - \varepsilon}{(m+1) \binom{m}{|F|}}, \frac{1 + \varepsilon}{(m+1) \binom{m}{|F|}} \right]. \quad (1)$$

Then both endpoint weights are at most $C/[m \min\{m, H\}]$, where $H = \log(1/\varepsilon)$. For dice, the event specifying which dice exceed the distinguished die is a union of strict orders, so (1) follows by summation. Equal row weights then prove Theorem 1.

Put $u_j = 1 - p_j$, $s = \sum_j u_j$, $\ell = \lfloor m/4 \rfloor$, $N = m - \ell$, $\alpha = \ell/m$, $x = \alpha s$, and $\theta = 1/16$. For a uniform ℓ -subset B , let $A_B = \prod_{j \in B} p_j$, $\bar{A} = \mathbb{E}_B A_B$, and $v_B = N^{-1} \sum_{j \notin B} \ell u_j$. The positive measure $\mu = (\ell + 1)w\bar{A}$ has mass in $[1 - \varepsilon, 1 + \varepsilon]$; it is not normalized to probability. Let $\nu = mw$ on the coordinate x , and let β have density

$$b_\ell(z) = \frac{\ell + 1}{\ell} (1 - z/\ell)^\ell \mathbf{1}_{[0, \ell]}(z).$$

Lemma 1 (Robust finite balance). *There is an absolute $\varepsilon_0 > 0$ such that, for $\varepsilon \leq \varepsilon_0$, the finite endpoint inputs hold with absolute constants:*

$$\max_j u_j \leq C(s+1)/m, \quad (2)$$

$$\Pr_w(s \leq S) \leq C(S+1)/m \quad (S \geq 0), \quad (3)$$

$$\mathbb{E}_w(s+1)^k e^{-as} \leq C_a^{k+1} k!/m \quad (k \geq 0, a > 0). \quad (4)$$

On $s \leq c_0 m$, all $p_j \geq 1/2$, for an absolute $c_0 > 0$ and sufficiently large m .

Proof. Summing (1) over unspecified coordinates shows that every partial pattern with a successes and b failures has probability within relative ε of $\int_0^1 t^a (1-t)^b dt$. This observation will also be used below.

Here are the quantitative changes to `finite_endpoint_odds_balance.tex`. Normalize each ℓ -failure tilt by its actual mass, which is in $[1-\varepsilon, 1+\varepsilon]$. All its undivided generating coefficients, including the coefficients with one specified derivative, then lie between the exact values times $(1-\varepsilon)/(1+\varepsilon)$ and their reciprocals. At a target row with $s \leq m/256$, fix i and choose $J \subseteq [m] \setminus \{i\}$, $|J| = \lceil 8s \rceil + 4$, containing all other columns with $p_j \leq 1/2$. Omit i from the additional failure-generating factors, as in that proof. The remaining target odds sum is at most $2s \leq |J|/4$. For ε_0 sufficiently small, the same coefficientwise bad-prefix bound with odds threshold one gives, for the good-row remaining odds variable Z ,

$$\mathbb{E}Z \geq \frac{3}{4}(|J|+1), \quad \mathbb{E}Z^2 \leq 2(|J|+1)(|J|+3).$$

Cauchy–Schwarz therefore gives probability at least $1/32$ on rows strictly before the target. The specified odds expectation is at most $2(|J|+1)/(m-|J|)$. As before, a zero target p_i is impossible on that prefix, and monotonicity yields $r_i \leq C(s+1)/m$. For larger s , (2) follows from $u_i \leq 1$. Choosing a smaller absolute c_0 gives $p_j \geq 1/2$ on $s \leq c_0 m$.

For (3), on $s \leq S \leq c_0 m$ choose $k = \lfloor cm/(S+1) \rfloor$ with a sufficiently small absolute c . Every k -column success product is bounded below absolutely on that suffix, while its expectation is at most $(1+\varepsilon)/(k+1)$. This proves the mass bound; larger S follow from total mass one. Laplace integration and a supremum bound for $(s+1)^k e^{-as/2}$ prove (4). All constants are uniform in ε . $\square$

Lemma 2 (The source error has no dimension factor). *For a polynomial $P(z) = \sum_{k=0}^D a_k z^k$, $D \leq N$,*

$$|(\ell+1)\mathbb{E}_w \mathbb{E}_B[A_B \text{Pol}_N(P)((\ell u_j)_{j \notin B})] - \beta(P)| \leq \varepsilon \sum_k |a_k| k!. \quad (3)$$

For the analogous marked measure with one fixed failure outside B , normalization $(\ell + 1)(\ell + 2)$ and scalar density proportional to $z(1 - z/\ell)^\ell$, the bound is $\varepsilon \sum_k |a_k|(k + 1)!$. If $P = Q_1 Q_2$ and the degree- d sums $Q_b = \sum_{j \leq d} c_{bj} L_j(\theta z)$ have coefficient sums B_b , these errors are at most $\varepsilon B_1 B_2 C_1^d$, for an absolute $C_1 > 1$.

Proof. A degree- k polarized monomial uses k distinct failures disjoint from the ℓ forced successes. Its exact normalized moment is

$$(\ell + 1)\ell^k \int_0^1 t^\ell (1 - t)^k dt \leq k!.$$

The marked version has one more failure and its moment is at most $(k + 1)!$. The partial-pattern observation in Lemma 1 gives the relative error term by term, proving (3). Finally, if $\|P\|_* = \sum |a_k|k!$, the explicit Laguerre coefficients give

$$\|L_j(\theta z) L_k(\theta z)\|_* \leq \sum_{a,b} \binom{j}{a} \binom{k}{b} 2^{a+b} \leq 3^{j+k},$$

since $(a + b)!/(a!b!) \leq 2^{a+b}$ and $\theta \leq 1$. The marked norm costs only an additional factor $2d + 1$, absorbed by increasing C_1 . Constants and degree zero are also covered by increasing the prefactor when necessary. $\square$

Lemma 3 (Robust Gaussian comparison). *For $\varepsilon \leq \varepsilon_0$, $d \leq c_1 m$ with an absolute $c_1 > 0$, and the Laguerre products of Lemma 2,*

$$|\mu(Q_1 Q_2) - \beta(Q_1 Q_2)| \leq C B_1 B_2 \{d/m + e^{-cm} + \varepsilon C_1^d\}. \quad (4)$$

The same assertion holds for the marked comparison after deletion of one column. All constants are uniform in that column.

Proof. The analytic comparison in `../power_per_die_lower_bound.tex` uses only the finite bounds (2)–(4), the algebraic success-weighted subset identities, Maclaurin’s inequality, and the Gaussian Laguerre derivative and centered-variance estimates. All these inputs remain valid. Split the subset comparison at $s = c_0 m$ rather than its former numerical cutoff. Its polarization series is bounded by $C B_1 B_2 \sum_{k \geq 2} (C \sqrt{d/m})^k (k + 1)^C$, so taking c_1 sufficiently small gives $C B_1 B_2 d/m$ even through linear degrees. Lemma 2 adds the displayed source error.

For the marked measure, deletion leaves an approximate tensor of $m - 1$ columns by the partial-pattern observation. The original rough bound

rearranges to $u_i \leq C(s_{-i} + 1)/m$. Thus its normalization $O(m^2)$ gives weighted factorial moments $C^{k+2}(k+1)!$, rather than $C^{k+1}k!$. This adds only a polynomial factor to the same geometric polarization series. The marked source error is precisely the second assertion of Lemma 2. No probability normalization of μ is needed in this analytic argument. $\square$

We retain the endpoint tests

$$T_d(y) = \sum_{j < d} (1-j/d)L_j(y), \quad U_d(z) = d^{-2}T_d(\theta z)^2, \quad R_d(z) = (2/(\theta d) - z)T_d(\theta z)^2.$$

Their reviewed scalar properties are

$$\beta(R_d) \geq c > 0, \quad \beta(U_d) \leq C/d, \quad U_d(z) \geq c \quad (0 \leq z \leq c_2/d). \quad (5)$$

The coefficient product norm is $O(1)$ for U_d and $O(d)$ for R_d , using $yT_d = d^{-1} \sum_{j < d} L_j - L_d$.

Lemma 4 (High-precision inputs). *Suppose $H \geq \sqrt{m}$. Then, for sufficiently large m ,*

$$\max_j u_j \leq C(s + \delta)/m, \quad \delta = m^{-1/3}, \quad (5)$$

$$\nu([0, t]) \leq C(t + h), \quad h = \max\{H^{-1}, m^{-3/4}\}. \quad (6)$$

For $\varepsilon = 0$, take $H^{-1} = 0$.

Proof. First (4), applied to U_d with $d \leq c_3\sqrt{m}$, gives $\mu(U_d) \leq C/d + Cd/m + \varepsilon C_1^d$. Choose $c_3 > 0$ small enough that $\varepsilon C_1^d \leq e^{-H/2}$. Near-zero positivity in (5), $\bar{A} \geq 1 - s$, and degree selection therefore give the initial CDF bound $\nu([0, t]) \leq C(t + m^{-1/2})$.

For the relative balance, repeat the marked Laplace argument in `quantitative_relative_endpoint_balance.tex`. Truncate the geometric Laguerre expansion of $e^{-tz/2}$ at $D = \lfloor c_3\sqrt{m} \rfloor$, reducing c_3 further if needed. Its coefficients $c_j = (1 - r)r^j$, $r = t/(t + 2\theta)$, have sum at most one and weighted degree sum $O(t)$. Applying (4) entrywise to $L_j L_k$ therefore gives, for either marked or unmarked measure,

$$|\mu_a(e^{-tx}) - \beta_a(e^{-tz})| \leq Ct/m + Ce^{-c\sqrt{m}/t} + Ce^{-H/2}. \quad (6)$$

This is uniform for $1/2 \leq t \leq m^{1/3}$. The source errors were bounded by their maximum εC_1^D , while the analytic errors use the weighted degree sum. Both measures may have masses $1 \pm \varepsilon$; positivity and the displayed

estimates suffice. Their Laplace bounds and the unmarked CDF bound give fixed constants a, b, c, C with $\mu_0(a/t < x < b/t) \geq c/t$ and $\mu_1(x \leq b/t) \leq C/t^2$, exactly as in that note. The unchanged monotone density relation $d\mu_1 = (\ell + 2)u_i d\mu_0$ proves (5).

For (6), use the elementary all-orders U_d comparison in `../size_bounds/elementary_endpoint_cdf_bootstrap.tex`. Its analytic proof uses only the rough balance and initial CDF; by Lemma 2 the approximate version is

$$|\mu(U_d) - \beta(U_d)| \leq C\{m^{-1} + m^{-1/2}d/m + \log d/(dm)\} + O(m^{-3}) + \varepsilon C_1^d. \quad (7)$$

This holds for $m^{1/4} \leq d \leq c_4 \min(H, m^{3/4})$, with c_4 a sufficiently small absolute constant. The source term is at most $e^{-H/2}$, and the right side is $O(1/d)$. The usual near-zero positivity and degree selection give $\nu([0, t]) \leq C(t + 1/D_{\max})$, $D_{\max} = c_4 \min(H, m^{3/4})$. This is (6). $\square$

Proof of the tensor bound and Theorem 1. The previously reviewed approximate linear lower bound, proved in `approximate_per_die_lower_bound.tex`, covers bounded H with endpoint weights $O(1/m)$, uniformly for $\varepsilon \leq 1/2$. Thus assume $H \geq H_0$, with H_0 a sufficiently large absolute constant, so $\varepsilon \leq \varepsilon_0$. Small bounded m are covered by enlarging constants.

If $H \leq \sqrt{m}$, choose $d = \lfloor c_5 H \rfloor$. Equations (4)–(5) bound the R_d error by

$$Cd^2/m + Cde^{-cm} + C\varepsilon dC_1^d.$$

First choose c_5 small, then H_0 large. The first term is at most Cc_5^2 , and the source is at most $Cde^{-H/2}$, so the error is less than half the positive scalar expectation. Therefore $x_K < 2/(\theta d)$. Apply U_r , $r = \lfloor d/16 \rfloor$; it is bounded below at x_K , and its comparison error is at most $Cr/m + Ce^{-cm} + Ce^{-H/2} = O(1/d)$. Hence $\mu(K) \leq C/d$. Since $s_K = O(1/d)$, $\bar{A}(K) \geq 1/2$ after increasing H_0 , and $w_K \leq C/(md) \leq C/(mH)$.

Now let $H \geq \sqrt{m}$, and use Lemma 4. Choose $d = \lfloor c_6 \min(H, m) \rfloor$, with a sufficiently small fixed $c_6 > 0$. We record explicitly the uniform estimates obtained from `../size_bounds/quadratic_per_die_lower_bound.tex` for this possibly sublinear degree. Its all-orders Gaussian argument, using (5)–(6), gives

$$\begin{aligned} |\mu(R_d) - \beta(R_d)| &\leq C \frac{d}{m} (1 + \delta \log(2d) + \delta h d) + o(1) + C\varepsilon dC_1^d, \\ |\mu(U_d) - \beta(U_d)| &\leq C \left\{ \frac{\log(2d)}{dm} + \frac{h}{m} + \frac{\delta}{m} + \frac{\delta h d}{m} \right\} + o(m^{-2}) + C\varepsilon C_1^d. \end{aligned} \quad (8)$$

Here the little-oh tails are uniform for $c\sqrt{m} \leq d \leq c_6 m$. To verify this without relying on a fixed proportional degree, the small- x errors are respectively $C(d^2/m)(1/d + \delta)(1/d + h)$ and $C(d/m)(1/d + \delta)(1/d + h)$, absorbed in (8). The middle-region envelopes are $C(d/m)e^{-cx}(1 + \delta/x)$ and $C(dm)^{-1}e^{-cx}(x^{-1} + \delta x^{-2})$. Their CDF integrals are precisely the terms in (8). Bad subsets have exponent at least $cm/(d\delta) \geq cm^{1/3}$; the outer region has exponent at least $cd \geq c\sqrt{m}$. Choose c_6 small enough for the same exponential margins as in the exact proof. Lemma 2 supplies the only additional terms, the displayed source errors.

In this regime

$$\delta h d \leq C m^{-1/3} \max\{1, m^{1/4}\} = O(m^{-1/12}), \quad \varepsilon C_1^d \leq e^{-H/2}.$$

Thus the first line of (8) is at most $Cc_6 + o(1)$, made smaller than half the scalar positivity constant. Again $x_K < 2/(\theta d)$. For $r = \lfloor d/16 \rfloor$, the second line of (8) is $o(1/d)$: after multiplication by d , every displayed term tends to zero. Since $U_r(x_K) \geq c$ and $\beta(U_r) \leq C/r$, we obtain $\mu(K) \leq C/d$. The same endpoint conversion gives

$$w_K \leq C/(md) \leq C/[m \min(H, m)].$$

Reversing rows and complementing all columns preserves the approximate pattern hypothesis and gives the same result for w_1 . The tensor-to-dice reduction at the start completes the proof. $\square$